

Calculus 1 Problem Templates for Dynamic Generation

Master Reference Sheet for Generator Patterns

Calculus Practice System (v.2026.01)

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Instructions for Professor: This document collects representative problem archetypes across all 7 core categories in the Calculus 1 practice hub. Please write your step-by-step gold-standard solution and LaTeX layout inside the `solutionbox` for each problem type. Once completed, the system will use your exact structure as the universal dynamic template for all randomized problem instances.

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1 Limits of Functions

1.1 Type 1.1: Factoring (0/0 Quadratic / Difference of Squares)

Problem 1.1: Limit by Factoring

Evaluate the limit:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{2^2 - 4}{2 - 2} = \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Factor the numerator using the difference of squares identity $a^2 - b^2 = (a - b)(a + b)$:

$$x^2 - 4 = (x - 2)(x + 2).$$

Step 3. Simplify the rational function by cancelling the common non-zero factor $(x - 2)$ for $x \neq 2$:

$$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2.$$

Step 4. Evaluate the limit by direct substitution:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 2 + 2 = 4.$$

1.2 Type 1.2: Square Root Conjugate (0/0)

Problem 1.2: Limit by Conjugate (Square Root)

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt{0+4} - 2}{0} = \frac{2 - 2}{0} = \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Multiply the numerator and denominator by the conjugate $(\sqrt{x+4} + 2)$:

$$\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \frac{(\sqrt{x+4})^2 - 2^2}{x(\sqrt{x+4} + 2)}.$$

Step 3. Simplify the expression:

$$\frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{\sqrt{0+4} + 2} = \frac{1}{2+2} = \frac{1}{4}.$$

1.3 Type 1.3: Cube Root Conjugate / Difference of Cubes (0/0)

Problem 1.3: Limit by Cube Root Conjugate

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+8} - 2}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt[3]{0+8} - 2}{0} = \frac{2 - 2}{0} = \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Use the identity $(a-b)(a^2+ab+b^2) = a^3-b^3$. Multiply numerator and denominator by $(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4$:

$$\frac{\sqrt[3]{x+8} - 2}{x} \cdot \frac{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4}{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4} = \frac{(x+8) - 8}{x((\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4)}.$$

Step 3. Simplify the expression:

$$\frac{x}{x((\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4)} = \frac{1}{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x \rightarrow 0} \frac{1}{(\sqrt[3]{8})^2 + 2\sqrt[3]{8} + 4} = \frac{1}{4 + 4 + 4} = \frac{1}{12}.$$

1.4 Type 1.4: Complex Fraction (0/0)

Problem 1.4: Limit of Complex Fractions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\frac{1}{x+3} - \frac{1}{3}}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\frac{1}{0+3} - \frac{1}{3}}{0} = \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Combine the terms in the numerator with a common denominator $3(x+3)$:

$$\frac{1}{x+3} - \frac{1}{3} = \frac{3 - (x+3)}{3(x+3)} = \frac{-x}{3(x+3)}.$$

Step 3. Simplify the complex fraction:

$$\frac{\frac{-x}{3(x+3)}}{x} = \frac{-x}{3x(x+3)} = -\frac{1}{3(x+3)}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x \rightarrow 0} -\frac{1}{3(x+3)} = -\frac{1}{3(0+3)} = -\frac{1}{9}.$$

1.5 Type 1.5: Double Conjugate (Numerator & Denominator)

Problem 1.5: Double Conjugate Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sqrt{x+9} - 3}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt{4} - 2}{\sqrt{9} - 3} = \frac{2 - 2}{3 - 3} = \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Multiply by both conjugates $(\sqrt{x+4} + 2)$ and $(\sqrt{x+9} + 3)$:

$$\frac{\sqrt{x+4} - 2}{\sqrt{x+9} - 3} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} \cdot \frac{\sqrt{x+9} + 3}{\sqrt{x+9} + 3} = \frac{((x+4) - 4)(\sqrt{x+9} + 3)}{((x+9) - 9)(\sqrt{x+4} + 2)}.$$

Step 3. Simplify and cancel common factor x :

$$\frac{x(\sqrt{x+9} + 3)}{x(\sqrt{x+4} + 2)} = \frac{\sqrt{x+9} + 3}{\sqrt{x+4} + 2}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+9} + 3}{\sqrt{x+4} + 2} = \frac{\sqrt{9} + 3}{\sqrt{4} + 2} = \frac{3 + 3}{2 + 2} = \frac{6}{4} = \frac{3}{2}.$$

1.6 Type 1.6: Basic Trigonometric Limits ($\frac{\sin kx}{x}$, $\frac{\tan kx}{\sin mx}$)

Problem 1.6: Trigonometric Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(5x)}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and standard trigonometric limits:

$$\begin{aligned}\lim_{u \rightarrow 0} \frac{\sin(u)}{u} &= 1 \\ \lim_{u \rightarrow 0} \frac{\tan(u)}{u} &= 1\end{aligned}$$

Step 2. Multiply and divide by $3x$ and $5x$ to obtain standard limit forms:

$$\frac{\tan(3x)}{\sin(5x)} = \frac{\frac{\tan(3x)}{3x} \cdot 3x}{\frac{\sin(5x)}{5x} \cdot 5x} = \frac{3}{5} \cdot \frac{\frac{\tan(3x)}{3x}}{\frac{\sin(5x)}{5x}}.$$

Step 3. Evaluate the limit as $x \rightarrow 0$:

$$\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(5x)} = \frac{3}{5} \cdot \frac{\lim_{x \rightarrow 0} \frac{\tan(3x)}{3x}}{\lim_{x \rightarrow 0} \frac{\sin(5x)}{5x}} = \frac{3}{5} \cdot \frac{1}{1} = \frac{3}{5}.$$

1.7 Type 1.7: Limits at Infinity (∞/∞ & $\infty - \infty$)

Problem 1.7: Limit at Infinity

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 4x} - x \right)$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form: $\infty - \infty$.

Step 2. Multiply and divide by the conjugate ($\sqrt{x^2 + 4x} + x$):

$$\frac{(\sqrt{x^2 + 4x} - x)(\sqrt{x^2 + 4x} + x)}{\sqrt{x^2 + 4x} + x} = \frac{(x^2 + 4x) - x^2}{\sqrt{x^2 + 4x} + x} = \frac{4x}{\sqrt{x^2 + 4x} + x}.$$

Step 3. Divide numerator and denominator by x (noting $\sqrt{x^2} = x$ for $x > 0$):

$$\frac{\frac{4x}{x}}{\sqrt{\frac{x^2 + 4x}{x^2}} + \frac{x}{x}} = \frac{4}{\sqrt{1 + \frac{4}{x}} + 1}.$$

Step 4. Evaluate the limit as $x \rightarrow \infty$, since $\lim_{x \rightarrow \infty} \frac{4}{x} = 0$:

$$\lim_{x \rightarrow \infty} \frac{4}{\sqrt{1 + \frac{4}{x}} + 1} = \frac{4}{\sqrt{1 + 0} + 1} = \frac{4}{2} = 2.$$

1.8 Type 1.8: One-Sided Limit with Absolute Value

Problem 1.8: One-Sided Limit with $|x|$

Evaluate the one-sided limit:

$$\lim_{x \rightarrow 2^-} \frac{|x - 2|}{x^2 - 4}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Apply the piecewise definition of the absolute value function:

$$|x - 2| = \begin{cases} x - 2, & \text{if } x \geq 2 \\ -(x - 2), & \text{if } x < 2 \end{cases}$$

Since $x \rightarrow 2^-$, we have $x < 2$, which implies $|x - 2| = -(x - 2)$.

Step 2. Factor the denominator: $x^2 - 4 = (x - 2)(x + 2)$.

Step 3. Substitute and simplify the expression for $x < 2$:

$$\frac{|x - 2|}{x^2 - 4} = \frac{-(x - 2)}{(x - 2)(x + 2)} = -\frac{1}{x + 2}.$$

Step 4. Evaluate the one-sided limit:

$$\lim_{x \rightarrow 2^-} \frac{|x - 2|}{x^2 - 4} = \lim_{x \rightarrow 2^-} \left(-\frac{1}{x + 2} \right) = -\frac{1}{2 + 2} = -\frac{1}{4}.$$

1.9 Type 1.9: L'Hôpital's Rule ($0/0$ or ∞/∞)

Problem 1.9: L'Hôpital's Rule

Evaluate the limit using L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin(3x)}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Verify the indeterminate form:

$$\lim_{x \rightarrow 0} (e^{2x} - 1) = 1 - 1 = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} \sin(3x) = 0 \implies \frac{0}{0} \quad (\text{Indeterminate Form}).$$

Step 2. Apply L'Hôpital's Rule by differentiating numerator and denominator:

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin(3x)} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}[e^{2x} - 1]}{\frac{d}{dx}[\sin(3x)]}.$$

Step 3. Calculate the derivatives:

$$\bullet \quad \frac{d}{dx}[e^{2x} - 1] = 2e^{2x}.$$

- $\frac{d}{dx}[\sin(3x)] = 3 \cos(3x)$.

Step 4. Evaluate the resulting limit:

$$\lim_{x \rightarrow 0} \frac{2e^{2x}}{3 \cos(3x)} = \frac{2e^0}{3 \cos(0)} = \frac{2(1)}{3(1)} = \frac{2}{3}.$$

2 Basic Derivatives & Term-by-Term Differentiation

2.1 Type 2.1: Polynomial with Fractional and Negative Powers

Problem 2.1: Powers and Radicals

Given $f(x) = 3x^3 + \frac{3}{x} + 5\sqrt{x}$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}u^n &= \frac{1}{u^{-n}} \\ \sqrt[n]{u^m} &= u^{\frac{m}{n}} \\ [u^n]' &= nu^{n-1}u' \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply linearity of differentiation:

$$f'(x) = \frac{d}{dx} \left(3x^3 + \frac{3}{x} + 5\sqrt{x} \right) = 3 \frac{d}{dx} x^3 + 3 \frac{d}{dx} \frac{1}{x} + 5 \frac{d}{dx} \sqrt{x}. \quad (1)$$

Step 3. Differentiate each term separately:

- Consider 1st term:

$$\frac{d}{dx} x^3 = 3x^2 \frac{d}{dx} x = 3x^2(1) = 3x^2.$$

- Consider 2nd term:

$$\frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = (-1)x^{-2} \frac{d}{dx} x = (-1)x^{-2}(1) = -\frac{1}{x^2}.$$

- Consider 3rd term:

$$\frac{d}{dx} \sqrt{x} = \frac{d}{dx} x^{\frac{1}{2}} = \frac{1}{2} \cdot x^{-\frac{1}{2}} \frac{d}{dx} x = \frac{1}{2} \cdot x^{-\frac{1}{2}}(1) = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}.$$

Step 4. Combine these three terms together, we obtain:

$$\begin{aligned}f'(x) &= 3 \frac{d}{dx} x^3 + 3 \frac{d}{dx} \frac{1}{x} + 5 \frac{d}{dx} \sqrt{x} \\ &= 3(3x^2) + 3 \left(-\frac{1}{x^2} \right) + 5 \left(\frac{1}{2\sqrt{x}} \right) \\ &= 9x^2 - \frac{3}{x^2} + \frac{5}{2\sqrt{x}}.\end{aligned}$$

Step 5. Evaluate at $x = 1$, we obtain:

$$f'(1) = 9(1)^2 - \frac{3}{(1)^2} + \frac{5}{2\sqrt{1}} = 9 - 3 + \frac{5}{2} = 6 + \frac{5}{2} = \frac{17}{2}.$$

2.2 Type 2.2: Basic Trigonometric Functions (Sine, Cosine, Secant, Tangent)

Problem 2.2: Trig Derivatives

Given $f(x) = 4 \sin(3x) - 2 \cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\sin(u)]' &= \cos(u)u' \\ [\cos(u)]' &= -\sin(u)u' \\ [u^n]' &= nu^{n-1}u' \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply linearity of differentiation:

$$f'(x) = \frac{d}{dx} (4 \sin(3x) - 2 \cos(3x)) = 4 \frac{d}{dx} \sin(3x) - 2 \frac{d}{dx} \cos(3x). \quad (2)$$

Step 3. Differentiate each term separately:

- Consider 1st term:

$$\frac{d}{dx} \sin(3x) = \cos(3x) \frac{d}{dx} (3x) = \cos(3x) \cdot 3 \frac{d}{dx} x = \cos(3x) \cdot 3(1) = 3 \cos(3x).$$

- Consider 2nd term:

$$\frac{d}{dx} \cos(3x) = -\sin(3x) \frac{d}{dx} (3x) = -\sin(3x) \cdot 3 \frac{d}{dx} x = -\sin(3x) \cdot 3(1) = -3 \sin(3x).$$

Step 4. Combine these two terms together, we obtain:

$$\begin{aligned}f'(x) &= 4 \frac{d}{dx} \sin(3x) - 2 \frac{d}{dx} \cos(3x) \\ &= 4(3 \cos(3x)) - 2(-3 \sin(3x)) \\ &= 12 \cos(3x) + 6 \sin(3x).\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $\cos(0) = 1$ and $\sin(0) = 0$, we obtain:

$$f'(0) = 12 \cos(0) + 6 \sin(0) = 12(1) + 6(0) = 12.$$

2.3 Type 2.3: Natural and General Exponential Functions (e^{kx} and a^u)

Problem 2.3: Exponential Derivatives

Given $f(x) = 3^{2x+1} + 2e^{3x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[a^u]' &= a^u \ln(a) u' \\ [e^u]' &= e^u u' \\ [u^n]' &= n u^{n-1} u' \\ [c]' &= 0 \\ (k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'\end{aligned}$$

Step 2. Apply linearity of differentiation:

$$f'(x) = \frac{d}{dx} (3^{2x+1} + 2e^{3x}) = \frac{d}{dx} 3^{2x+1} + 2 \frac{d}{dx} e^{3x}. \quad (3)$$

Step 3. Differentiate each term separately:

- Consider 1st term:

$$\frac{d}{dx} 3^{2x+1} = 3^{2x+1} \ln(3) \frac{d}{dx} (2x+1) = 3^{2x+1} \ln(3) \left(2 \frac{d}{dx} x + \frac{d}{dx} (1) \right) = 3^{2x+1} \ln(3) (2(1)+0) = 2 \ln(3)$$

- Consider 2nd term:

$$\frac{d}{dx} e^{3x} = e^{3x} \frac{d}{dx} (3x) = e^{3x} \cdot 3 \frac{d}{dx} x = e^{3x} \cdot 3(1) = 3e^{3x}.$$

Step 4. Combine these two terms together, we obtain:

$$\begin{aligned}f'(x) &= \frac{d}{dx} 3^{2x+1} + 2 \frac{d}{dx} e^{3x} \\ &= 2 \ln(3) \cdot 3^{2x+1} + 2(3e^{3x}) \\ &= 2 \ln(3) \cdot 3^{2x+1} + 6e^{3x}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $3^{2(0)+1} = 3^1 = 3$ and $e^{3(0)} = e^0 = 1$, we obtain:

$$\begin{aligned}f'(0) &= 2 \ln(3) \cdot 3^1 + 6(1) \\ &= 6 \ln(3) + 6 = 6(\ln(3) + 1).\end{aligned}$$

2.4 Type 2.4: Sum of Inverse Trigonometric Functions

Problem 2.4: Inverse Trig Derivatives

Given $f(x) = 3 \arccos(4x) + 2 \operatorname{arccot}(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\arccos(u)]' &= -\frac{1}{\sqrt{1-u^2}}u' \\ [\operatorname{arccot}(u)]' &= -\frac{1}{1+u^2}u' \\ [u^n]' &= nu^{n-1}u' \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply linearity of differentiation:

$$f'(x) = \frac{d}{dx} (3 \arccos(4x) + 2 \operatorname{arccot}(x)) = 3 \frac{d}{dx} \arccos(4x) + 2 \frac{d}{dx} \operatorname{arccot}(x). \quad (4)$$

Step 3. Differentiate each term separately:

- Consider 1st term:

$$\frac{d}{dx} \arccos(4x) = -\frac{1}{\sqrt{1-(4x)^2}} \frac{d}{dx} (4x) = -\frac{1}{\sqrt{1-16x^2}} \cdot 4 \frac{d}{dx} x = -\frac{1}{\sqrt{1-16x^2}} \cdot 4(1) = -\frac{4}{\sqrt{1-16x^2}}$$

- Consider 2nd term:

$$\frac{d}{dx} \operatorname{arccot}(x) = -\frac{1}{1+x^2} \frac{d}{dx} x = -\frac{1}{1+x^2} (1) = -\frac{1}{1+x^2}.$$

Step 4. Combine these two terms together, we obtain:

$$\begin{aligned}f'(x) &= 3 \frac{d}{dx} \arccos(4x) + 2 \frac{d}{dx} \operatorname{arccot}(x) \\ &= 3 \left(-\frac{4}{\sqrt{1-16x^2}} \right) + 2 \left(-\frac{1}{1+x^2} \right) \\ &= -\frac{12}{\sqrt{1-16x^2}} - \frac{2}{1+x^2}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = -\frac{12}{\sqrt{1-0}} - \frac{2}{1+0} = -12 - 2 = -14.$$

3 Product Rule: $(uv)' = uv' + vu'$

3.1 Type 3.1: Polynomial $\times$ Exponential

Problem 3.1: Product Rule (Poly $\times$ Exp)

Given $f(x) = (2x + 3)e^{2x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}(u \cdot v)' &= uv' + vu' \\ [e^u]' &= e^u u' \\ [u^n]' &= nu^{n-1}u' \\ [c]' &= 0 \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply the product rule:

$$f'(x) = \frac{d}{dx} [(2x + 3)e^{2x}] = (2x + 3) \frac{d}{dx} [e^{2x}] + e^{2x} \frac{d}{dx} [2x + 3]. \quad (5)$$

Step 3. Differentiate each factor separately:

- Consider the derivative of e^{2x} :

$$\frac{d}{dx} [e^{2x}] = e^{2x} \frac{d}{dx} (2x) = e^{2x} \cdot 2 \frac{d}{dx} x = e^{2x} \cdot 2(1) = 2e^{2x}.$$

- Consider the derivative of $(2x + 3)$:

$$\frac{d}{dx} [2x + 3] = 2 \frac{d}{dx} x + \frac{d}{dx} (3) = 2(1) + 0 = 2.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= (2x + 3) \frac{d}{dx} [e^{2x}] + e^{2x} \frac{d}{dx} [2x + 3] \\ &= (2x + 3)(2e^{2x}) + e^{2x}(2) \\ &= (4x + 6)e^{2x} + 2e^{2x} \\ &= (4x + 8)e^{2x}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $e^{2(0)} = e^0 = 1$, we obtain:

$$f'(0) = (4(0) + 8)e^0 = 8(1) = 8.$$

3.2 Type 3.2: Exponential $\times$ Trigonometric (Damped Oscillation)

Problem 3.2: Product Rule (Exp $\times$ Trig)

Given $f(x) = e^{-2x} \cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}(u \cdot v)' &= uv' + vu' \\ [e^u]' &= e^u u' \\ [\cos(u)]' &= -\sin(u)u' \\ [u^n]' &= nu^{n-1}u' \\ (k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'\end{aligned}$$

Step 2. Apply the product rule:

$$f'(x) = \frac{d}{dx} [e^{-2x} \cos(3x)] = e^{-2x} \frac{d}{dx} [\cos(3x)] + \cos(3x) \frac{d}{dx} [e^{-2x}]. \quad (6)$$

Step 3. Differentiate each factor separately:

- Consider the derivative of $\cos(3x)$:

$$\frac{d}{dx} [\cos(3x)] = -\sin(3x) \frac{d}{dx} (3x) = -\sin(3x) \cdot 3 \frac{d}{dx} x = -\sin(3x) \cdot 3(1) = -3 \sin(3x).$$

- Consider the derivative of e^{-2x} :

$$\frac{d}{dx} [e^{-2x}] = e^{-2x} \frac{d}{dx} (-2x) = e^{-2x} \cdot (-2) \frac{d}{dx} x = e^{-2x} \cdot (-2)(1) = -2e^{-2x}.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= e^{-2x} \frac{d}{dx} [\cos(3x)] + \cos(3x) \frac{d}{dx} [e^{-2x}] \\ &= e^{-2x} (-3 \sin(3x)) + \cos(3x) (-2e^{-2x}) \\ &= -3e^{-2x} \sin(3x) - 2e^{-2x} \cos(3x) \\ &= -(3 \sin(3x) + 2 \cos(3x))e^{-2x}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $e^0 = 1$, $\sin(0) = 0$, and $\cos(0) = 1$, we obtain:

$$f'(0) = -(3(0) + 2(1))e^0 = -2(1) = -2.$$

3.3 Type 3.3: Polynomial / Exponential $\times$ Inverse Trigonometric

Problem 3.3: Product Rule with Inverse Trig

Given $f(x) = (e^{3x} + 2x) \arctan(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}(u \cdot v)' &= uv' + vu' \\ [\arctan(u)]' &= \frac{1}{1+u^2}u' \\ [e^u]' &= e^u u' \\ [u^n]' &= nu^{n-1}u' \\ (k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'\end{aligned}$$

Step 2. Apply the product rule:

$$f'(x) = \frac{d}{dx} [(e^{3x} + 2x) \arctan(x)] = (e^{3x} + 2x) \frac{d}{dx} [\arctan(x)] + \arctan(x) \frac{d}{dx} [e^{3x} + 2x]. \quad (7)$$

Step 3. Differentiate each factor separately:

- Consider the derivative of $\arctan(x)$:

$$\frac{d}{dx} [\arctan(x)] = \frac{1}{1+x^2} \frac{d}{dx} x = \frac{1}{1+x^2} (1) = \frac{1}{1+x^2}.$$

- Consider the derivative of $(e^{3x} + 2x)$:

$$\frac{d}{dx} [e^{3x} + 2x] = \frac{d}{dx} [e^{3x}] + 2 \frac{d}{dx} x = e^{3x} \cdot 3(1) + 2(1) = 3e^{3x} + 2.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= (e^{3x} + 2x) \frac{d}{dx} [\arctan(x)] + \arctan(x) \frac{d}{dx} [e^{3x} + 2x] \\ &= (e^{3x} + 2x) \left(\frac{1}{1+x^2} \right) + \arctan(x) (3e^{3x} + 2) \\ &= \frac{e^{3x} + 2x}{1+x^2} + (3e^{3x} + 2) \arctan(x).\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $e^0 = 1$ and $\arctan(0) = 0$, we obtain:

$$f'(0) = \frac{1+0}{1+0} + (3(1) + 2) \arctan(0) = 1 + 5(0) = 1.$$

4 Quotient Rule: $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$

4.1 Type 4.1: Rational Polynomial Function

Problem 4.1: Quotient Rule (Polynomials)

Given $f(x) = \frac{2x+3}{x-4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}\left(\frac{u}{v}\right)' &= \frac{vu' - uv'}{v^2} \\ [u^n]' &= nu^{n-1}u' \\ [c]' &= 0 \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply the quotient rule:

$$f'(x) = \frac{d}{dx} \left[\frac{2x+3}{x-4} \right] = \frac{(x-4)\frac{d}{dx}[2x+3] - (2x+3)\frac{d}{dx}[x-4]}{(x-4)^2}. \quad (8)$$

Step 3. Differentiate numerator and denominator separately:

- Consider the derivative of the numerator:

$$\frac{d}{dx}[2x+3] = 2\frac{d}{dx}x + \frac{d}{dx}(3) = 2(1) + 0 = 2.$$

- Consider the derivative of the denominator:

$$\frac{d}{dx}[x-4] = \frac{d}{dx}x - \frac{d}{dx}(4) = 1(1) - 0 = 1.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= \frac{(x-4)(2) - (2x+3)(1)}{(x-4)^2} \\ &= \frac{2x-8-2x-3}{(x-4)^2} \\ &= -\frac{11}{(x-4)^2}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = -\frac{11}{(0-4)^2} = -\frac{11}{16}.$$

4.2 Type 4.2: Trigonometric / Radical Quotient

Problem 4.2: Quotient Rule (Trig / Radical)

Given $f(x) = \frac{\sin(2x)}{\sqrt{x^2+1}}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}\left(\frac{u}{v}\right)' &= \frac{vu' - uv'}{v^2} \\ [\sin(u)]' &= \cos(u)u' \\ [\sqrt{u}]' &= \frac{1}{2\sqrt{u}}u' \\ [u^n]' &= nu^{n-1}u' \\ [c]' &= 0\end{aligned}$$

Step 2. Apply the quotient rule:

$$f'(x) = \frac{d}{dx} \left[\frac{\sin(2x)}{\sqrt{x^2+1}} \right] = \frac{\sqrt{x^2+1} \frac{d}{dx} [\sin(2x)] - \sin(2x) \frac{d}{dx} [\sqrt{x^2+1}]}{\left(\sqrt{x^2+1}\right)^2}. \quad (9)$$

Step 3. Differentiate numerator and denominator separately:

- Consider the derivative of the numerator $\sin(2x)$:

$$\frac{d}{dx} [\sin(2x)] = \cos(2x) \frac{d}{dx} (2x) = \cos(2x) \cdot 2(1) = 2 \cos(2x).$$

- Consider the derivative of the denominator $\sqrt{x^2+1}$:

$$\frac{d}{dx} [\sqrt{x^2+1}] = \frac{1}{2\sqrt{x^2+1}} \frac{d}{dx} [x^2+1] = \frac{2x(1)+0}{2\sqrt{x^2+1}} = \frac{x}{\sqrt{x^2+1}}.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= \frac{\sqrt{x^2+1}(2 \cos(2x)) - \sin(2x) \left(\frac{x}{\sqrt{x^2+1}}\right)}{x^2+1} \\ &= \frac{2(x^2+1) \cos(2x) - x \sin(2x)}{(x^2+1)^{\frac{3}{2}}}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, since $\cos(0) = 1$ and $\sin(0) = 0$, we obtain:

$$f'(0) = \frac{2(0+1)(1) - 0(0)}{(0+1)^{\frac{3}{2}}} = \frac{2}{1} = 2.$$

5 Chain Rule & Composite Functions

5.1 Type 5.1: Radical with Quadratic Inside $\sqrt{u(x)}$

Problem 5.1: Chain Rule with Radical

Given $f(x) = \sqrt{2x^2 + 3x + 4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\sqrt{u}]' &= \frac{1}{2\sqrt{u}}u' \\[u^n]' &= nu^{n-1}u' \\[c]' &= 0 \\(k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply the chain rule by letting $u = 2x^2 + 3x + 4$:

$$f'(x) = \frac{d}{dx} [\sqrt{2x^2 + 3x + 4}] = \frac{1}{2\sqrt{2x^2 + 3x + 4}} \frac{d}{dx} [2x^2 + 3x + 4]. \quad (10)$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx} [2x^2 + 3x + 4] = 2\frac{d}{dx} [x^2] + 3\frac{d}{dx} x + \frac{d}{dx} (4) = 2(2x \cdot 1) + 3(1) + 0 = 4x + 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}f'(x) &= \frac{1}{2\sqrt{2x^2 + 3x + 4}} (4x + 3) \\&= \frac{4x + 3}{2\sqrt{2x^2 + 3x + 4}}.\end{aligned}$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = \frac{4(0) + 3}{2\sqrt{2(0)^2 + 3(0) + 4}} = \frac{3}{2\sqrt{4}} = \frac{3}{2(2)} = \frac{3}{4}.$$

5.2 Type 5.2: Logarithmic with Polynomial Inside $\ln(u(x))$

Problem 5.2: Chain Rule with Natural Log

Given $f(x) = \ln(x^2 + 3x + 2)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\ln(u)]' &= \frac{1}{u}u' \\ [u^n]' &= nu^{n-1}u' \\ [c]' &= 0 \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Apply the chain rule by letting $u = x^2 + 3x + 2$:

$$f'(x) = \frac{d}{dx} [\ln(x^2 + 3x + 2)] = \frac{1}{x^2 + 3x + 2} \frac{d}{dx} [x^2 + 3x + 2]. \quad (11)$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx} [x^2 + 3x + 2] = \frac{d}{dx} [x^2] + 3 \frac{d}{dx} x + \frac{d}{dx} (2) = 2x(1) + 3(1) + 0 = 2x + 3.$$

Step 4. Combine these terms together, we obtain:

$$f'(x) = \frac{2x + 3}{x^2 + 3x + 2}.$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = \frac{2(0) + 3}{0^2 + 3(0) + 2} = \frac{3}{2}.$$

5.3 Type 5.3: Trigonometric with Linear / Quadratic Inside**Problem 5.3: Chain Rule with Cotangent**

Given $f(x) = 2 \cot \left(3x + \frac{\pi}{4} \right)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\cot(u)]' &= -\csc^2(u)u' \\ [u^n]' &= nu^{n-1}u' \\ [c]' &= 0 \\ (k \cdot g)' &= k \cdot g'\end{aligned}$$

Step 2. Apply constant multiple and chain rule by letting $u = 3x + \frac{\pi}{4}$:

$$f'(x) = 2 \frac{d}{dx} \left[\cot \left(3x + \frac{\pi}{4} \right) \right] = 2 \left(-\csc^2 \left(3x + \frac{\pi}{4} \right) \frac{d}{dx} \left[3x + \frac{\pi}{4} \right] \right). \quad (12)$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx} \left[3x + \frac{\pi}{4} \right] = 3 \frac{d}{dx} x + \frac{d}{dx} \left(\frac{\pi}{4} \right) = 3(1) + 0 = 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= 2 \left(-\csc^2 \left(3x + \frac{\pi}{4} \right) \cdot 3 \right) \\ &= -6 \csc^2 \left(3x + \frac{\pi}{4} \right). \end{aligned}$$

Step 5. Evaluate at $x = 0$, since $\csc \left(\frac{\pi}{4} \right) = \sqrt{2}$, we obtain $\csc^2 \left(\frac{\pi}{4} \right) = 2$:

$$f'(0) = -6 \csc^2 \left(\frac{\pi}{4} \right) = -6(2) = -12.$$

5.4 Type 5.4: Inverse Trigonometric with Chain Rule

Problem 5.4: Chain Rule with ArcSec

Given $f(x) = 2 \operatorname{arcsec}(3x)$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned} [\operatorname{arcsec}(u)]' &= \frac{1}{|u|\sqrt{u^2 - 1}} u' \\ [u^n]' &= nu^{n-1}u' \\ (k \cdot g)' &= k \cdot g' \end{aligned}$$

Step 2. Apply constant multiple and chain rule by letting $u = 3x$:

$$f'(x) = 2 \frac{d}{dx} [\operatorname{arcsec}(3x)] = 2 \left(\frac{1}{|3x|\sqrt{(3x)^2 - 1}} \frac{d}{dx} [3x] \right). \quad (13)$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx} [3x] = 3 \frac{d}{dx} x = 3(1) = 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= 2 \left(\frac{1}{|3x|\sqrt{9x^2 - 1}} \cdot 3 \right) \\ &= \frac{6}{|3x|\sqrt{9x^2 - 1}} = \frac{2}{|x|\sqrt{9x^2 - 1}}. \end{aligned}$$

Step 5. Evaluate at $x = 1$, we obtain:

$$f'(1) = \frac{2}{|1|\sqrt{9(1)^2 - 1}} = \frac{2}{\sqrt{8}} = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

5.5 Type 5.5: Nested Composite Function (Trig(Inverse Trig(kx)))

Problem 5.5: Nested Trig of Inverse Trig

Given $f(x) = \tan(\arcsin(3x))$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}[\tan(u)]' &= \sec^2(u)u' \\ [\arcsin(u)]' &= \frac{1}{\sqrt{1-u^2}}u' \\ [u^n]' &= nu^{n-1}u'\end{aligned}$$

Step 2. Apply the chain rule from outer to inner functions:

$$f'(x) = \frac{d}{dx} [\tan(\arcsin(3x))] = \sec^2(\arcsin(3x)) \frac{d}{dx} [\arcsin(3x)]. \quad (14)$$

Step 3. Differentiate the inner function $\arcsin(3x)$:

$$\frac{d}{dx} [\arcsin(3x)] = \frac{1}{\sqrt{1-(3x)^2}} \frac{d}{dx} [3x] = \frac{1}{\sqrt{1-9x^2}} \cdot 3 \frac{d}{dx} x = \frac{3}{\sqrt{1-9x^2}}.$$

Step 4. Combine these terms together, we obtain:

$$f'(x) = \sec^2(\arcsin(3x)) \cdot \frac{3}{\sqrt{1-9x^2}} = \frac{3 \sec^2(\arcsin(3x))}{\sqrt{1-9x^2}}.$$

Step 5. Evaluate at $x = 0$, since $\arcsin(0) = 0$ and $\sec(0) = 1$, we obtain:

$$f'(0) = \frac{3 \sec^2(0)}{\sqrt{1-0}} = \frac{3(1)^2}{1} = 3.$$

5.6 Type 5.6: Logarithmic Differentiation ($y = x^{kx}$)

Problem 5.6: Logarithmic Differentiation

Given $f(x) = x^{3x}$ for $x > 0$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and properties used in logarithmic differentiation:

$$\begin{aligned}\ln(u^r) &= r \ln(u) \\ [\ln(y)]' &= \frac{1}{y} y' \\ (u \cdot v)' &= uv' + vu' \\ [\ln(x)]' &= \frac{1}{x}\end{aligned}$$

Step 2. Take natural logarithm of both sides and differentiate: Let $y = x^{3x}$. Then $\ln(y) = 3x \ln(x)$.

$$\begin{aligned}\frac{d}{dx}[\ln(y)] &= \frac{d}{dx}[3x \ln(x)] \\ \frac{1}{y} \frac{dy}{dx} &= 3x \frac{d}{dx}[\ln(x)] + \ln(x) \frac{d}{dx}[3x].\end{aligned}$$

Step 3. Differentiate each part on the right-hand side:

- $3x \frac{d}{dx}[\ln(x)] = 3x \left(\frac{1}{x}\right) = 3.$
- $\ln(x) \frac{d}{dx}[3x] = \ln(x) \cdot 3(1) = 3 \ln(x).$

Step 4. Combine and solve for $\frac{dy}{dx} = f'(x)$:

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= 3 + 3 \ln(x) = 3(1 + \ln(x)) \\ f'(x) &= y \cdot 3(1 + \ln(x)) \\ f'(x) &= 3x^{3x}(1 + \ln(x)).\end{aligned}$$

Step 5. Evaluate at $x = 1$, since $\ln(1) = 0$ and $1^3 = 1$, we obtain:

$$f'(1) = 3(1)^3(1 + \ln(1)) = 3(1)(1 + 0) = 3.$$

5.7 Type 5.7: Higher-Order Derivative ($f''(x)$)

Problem 5.7: Second Derivative

Given $f(x) = x^2 e^{2x}$, find $f''(x)$ and evaluate $f''(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}(u \cdot v)' &= uv' + vu' \\ [e^u]' &= e^u u' \\ [u^n]' &= nu^{n-1} u' \\ (k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'\end{aligned}$$

Step 2. Find the first derivative $f'(x)$ using the product rule:

$$f'(x) = \frac{d}{dx}[x^2 e^{2x}] = x^2 \frac{d}{dx}[e^{2x}] + e^{2x} \frac{d}{dx}[x^2]. \quad (15)$$

Step 3. Differentiate each component:

$$\begin{aligned}\frac{d}{dx}[e^{2x}] &= e^{2x} \frac{d}{dx}[2x] = 2e^{2x} \\ \frac{d}{dx}[x^2] &= 2x \frac{dx}{dx} = 2x(1) = 2x\end{aligned}$$

Step 4. Combine and simplify $f'(x)$:

$$f'(x) = x^2(2e^{2x}) + e^{2x}(2x) = 2x^2e^{2x} + 2xe^{2x} = (2x^2 + 2x)e^{2x}.$$

Step 5. Find the second derivative $f''(x)$ by differentiating $f'(x)$ again:

$$\begin{aligned} f''(x) &= \frac{d}{dx} [(2x^2 + 2x)e^{2x}] \\ &= (2x^2 + 2x) \frac{d}{dx} [e^{2x}] + e^{2x} \frac{d}{dx} [2x^2 + 2x] \\ &= (2x^2 + 2x)(2e^{2x}) + e^{2x}(4x + 2) \\ &= (4x^2 + 4x + 4x + 2)e^{2x} \\ &= (4x^2 + 8x + 2)e^{2x}. \end{aligned}$$

Evaluate at $x = 0$:

$$f''(0) = (4(0)^2 + 8(0) + 2)e^0 = 2(1) = 2.$$

6 Implicit Differentiation

6.1 Type 6.1: Circle Equation ($x^2 + y^2 = r^2$)

Problem 6.1: Implicit Differentiation of a Circle

Find $\frac{dy}{dx}$ for the equation $x^2 + y^2 = 25$ at the point $(3, 4)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}\frac{d}{dx}[x^n] &= nx^{n-1} \\ \frac{d}{dx}[y^n] &= ny^{n-1} \frac{dy}{dx} \\ \frac{d}{dx}[c] &= 0\end{aligned}$$

Step 2. Differentiate both sides of the equation with respect to x :

$$\begin{aligned}\frac{d}{dx}[x^2 + y^2] &= \frac{d}{dx}[25] \\ \frac{d}{dx}[x^2] + \frac{d}{dx}[y^2] &= 0.\end{aligned}$$

Step 3. Calculate each derivative term:

$$\begin{aligned}\frac{d}{dx}[x^2] &= 2x \frac{dx}{dx} = 2x(1) = 2x \\ \frac{d}{dx}[y^2] &= 2y \frac{dy}{dx}\end{aligned}$$

Step 4. Substitute back and isolate $\frac{dy}{dx}$:

$$\begin{aligned}2x + 2y \frac{dy}{dx} &= 0 \\ 2y \frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{2x}{2y} = -\frac{x}{y}.\end{aligned}$$

Step 5. Substitute the given point $(x, y) = (3, 4)$:

$$\left. \frac{dy}{dx} \right|_{(3,4)} = -\frac{3}{4}.$$

6.2 Type 6.2: Mixed Polynomial with xy Product Term

Problem 6.2: Implicit Differentiation with Product Rule

Find $\frac{dy}{dx}$ for the equation $x^3 + y^3 - 3xy = 5$ at the point (x_0, y_0) .

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}\frac{d}{dx}[u \cdot v] &= u \frac{dv}{dx} + v \frac{du}{dx} \\ \frac{d}{dx}[y^n] &= ny^{n-1} \frac{dy}{dx} \\ \frac{d}{dx}[x^n] &= nx^{n-1} \\ \frac{d}{dx}[c] &= 0\end{aligned}$$

Step 2. Differentiate both sides of the equation with respect to x :

$$\begin{aligned}\frac{d}{dx}[x^3 + y^3 - 3xy] &= \frac{d}{dx}[5] \\ \frac{d}{dx}[x^3] + \frac{d}{dx}[y^3] - 3 \frac{d}{dx}[xy] &= 0.\end{aligned}$$

Step 3. Calculate each derivative term:

$$\begin{aligned}\frac{d}{dx}[x^3] &= 3x^2 \frac{dx}{dx} = 3x^2(1) = 3x^2 \\ \frac{d}{dx}[y^3] &= 3y^2 \frac{dy}{dx} \\ \frac{d}{dx}[xy] &= x \frac{dy}{dx} + y \frac{dx}{dx} = x \frac{dy}{dx} + y(1) = x \frac{dy}{dx} + y\end{aligned}$$

Step 4. Combine and isolate $\frac{dy}{dx}$:

$$\begin{aligned}3x^2 + 3y^2 \frac{dy}{dx} - 3 \left(x \frac{dy}{dx} + y \right) &= 0 \\ 3x^2 + 3y^2 \frac{dy}{dx} - 3x \frac{dy}{dx} - 3y &= 0 \\ (3y^2 - 3x) \frac{dy}{dx} &= 3y - 3x^2 \\ \frac{dy}{dx} &= \frac{3y - 3x^2}{3y^2 - 3x} = \frac{y - x^2}{y^2 - x}.\end{aligned}$$

Step 5. Evaluate at the given point (x_0, y_0) , we obtain:

$$\left. \frac{dy}{dx} \right|_{(x_0, y_0)} = \frac{y_0 - x_0^2}{y_0^2 - x_0}.$$

7 Tangent & Normal Lines

7.1 Type 7.1: Equation of the Tangent Line

Problem 7.1: Tangent Line Equation

Find the slope of the tangent line m_t and the equation of the tangent line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Formulas for the tangent line:

$$\begin{aligned}m_t &= f'(x_0) \\ y - y_0 &= m_t(x - x_0)\end{aligned}$$

Step 2. Find the derivative $f'(x)$:

$$f'(x) = \frac{d}{dx} [x^2 - 3x + 2] = 2x - 3. \quad (16)$$

Step 3. Find the point of tangency (x_0, y_0) by evaluating $y(2)$:

$$y_0 = f(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$$

implies $(x_0, y_0) = (2, 0)$.

Step 4. Find the slope of the tangent line m_t :

$$m_t = f'(2) = 2(2) - 3 = 4 - 3 = 1.$$

Step 5. Construct the equation of the tangent line:

$$\begin{aligned}y - y_0 &= m_t(x - x_0) \\ y - 0 &= 1(x - 2) \\ y &= x - 2 \quad (\text{or } x - y - 2 = 0).\end{aligned}$$

7.2 Type 7.2: Equation of the Normal Line

Problem 7.2: Normal Line Equation

Find the slope of the normal line m_n and the equation of the normal line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Formulas for the normal line:

$$\begin{aligned}m_t &= f'(x_0) \\ m_n &= -\frac{1}{m_t} \quad (m_t \neq 0) \\ y - y_0 &= m_n(x - x_0)\end{aligned}$$

Step 2. Find the derivative $f'(x)$:

$$f'(x) = \frac{d}{dx} [x^2 - 3x + 2] = 2x - 3. \quad (17)$$

Step 3. Find the point (x_0, y_0) by evaluating $y(2)$:

$$y_0 = f(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$$

implies $(x_0, y_0) = (2, 0)$.

Step 4. Find the slope of the normal line m_n :

$$\begin{aligned} m_t &= f'(2) = 2(2) - 3 = 1 \\ m_n &= -\frac{1}{m_t} = -\frac{1}{1} = -1. \end{aligned}$$

Step 5. Construct the equation of the normal line:

$$\begin{aligned} y - y_0 &= m_n(x - x_0) \\ y - 0 &= -1(x - 2) \\ y &= -x + 2 \quad (\text{or } x + y - 2 = 0). \end{aligned}$$